Proposal Defense

Socratic AI: Agentic AI for Fostering Critical Thinking Education

A multi-agent tutor that asks instead of answers, and measures how well the learner reasons

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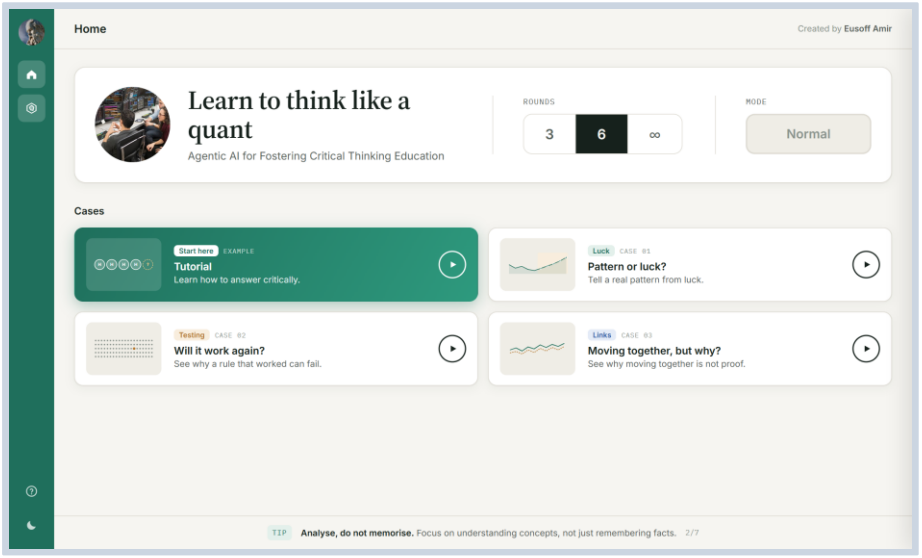
Course : EFB4042 Final Year Project I

Project code : 26S005

Date : 21st October 2026

OUTLINE OF PRESENTATION

1. Introduction
2. Problem Statement, Objectives, Scope of Study
3. Literature Review
4. Proposed Methodology
5. Gantt Chart & Milestones
6. Preliminary Results
7. Conclusion



The working prototype, live online

INTRODUCTION

Generative AI now gives students a finished answer in seconds, so they can skip the thinking.

62%

of students use generative AI at least weekly [12]

319

knowledge workers studied: more trust in AI meant less critical thinking [10]

4

costs of "outsourced reasoning": fewer ideas, less effort, weaker memory, less reflection [11]

The idea of this project

Turn the same AI around. Instead of an answer machine, build a tutor that asks Socratic questions in a set order, so the learner builds the understanding, and then measure the quality of the learner's reasoning against a framework.

PROBLEM STATEMENT

Students increasingly hand their thinking to generative AI, and studies link this to weaker critical thinking [10]. Socratic AI tutors can refuse to give answers, but they share three gaps.

1

No ladder

Questions only lead toward an answer, so learners are never pushed to ask whether they are asking the right question.

2

No measurement

They do not measure the learner's critical thinking against any framework.

3

No adaptation

They do not adapt to what each learner is missing.

Seen in early testing of two Gemini Gem tutors: *"bro i explained this before"*

As a result, neither the learner nor the lecturer can tell whether critical thinking is actually improving.

OBJECTIVES

1	**Teach**	Undergraduates new to a topic build their own reasoning, guided step by step up Paul's six levels of Socratic questioning, without being given the answers.
2	**Assess**	Each learner's critical thinking attainment is measured from the conversation against the Paul-Elder framework (the nine standards, plus the type of reasoning and knowledge used), with marks that agree with human markers.
3	**Adapt**	The support and difficulty match each learner's gaps, and the learner can see what they have mastered and what to work on next.

These match the three objectives and deliverables in the official project listing (26S005).

SCOPE OF STUDY

1

One quant topic

Telling a real pattern in data from luck, for beginners, with no complex machine learning. Three short cases.

2

Multi-agent tutor

Four AI agents (Assessor, Planner, Tutor, Checker) guarded by plain-code rules, on the free Gemini model.

3

Framework assessment

6 Socratic levels and 9 standards now, then reasoning type and knowledge type, with a learner dashboard.

Target learners: university students new to quant (early testers are Dr Ho's students).

Future work, outside FYP: a second topic, real reading material with retrieval (RAG) [15], knowledge graphs, and a full class study (needs a paid key and ethics approval).

LITERATURE REVIEW

Agentic AI Building Blocks

Technique	Year	Core idea	Use in this project	Ref
Prompting	2025	A prompt is an instruction plus context; output depends on how the context is built	Every agent gets a structured JSON prompt	[16]
Chain of Thought	2022	Break a task into intermediate reasoning steps	The Planner reasons before choosing the next move	[17]
ReAct	2023	Reason, act, observe, then repeat	The Tutor runs a simulation, reads the result, then replies	[18]
RAG	2023	Ground answers in retrieved knowledge (naive, advanced, modular)	Case notes now; real RAG and knowledge graphs are future work	[15], [19]
Model Context Protocol	2024	A standard way to give an LLM tools, resources and prompts	Future: query a knowledge graph and learner records	[20]
Multi-agent systems	2024	Specialised agents collaborate (MetaGPT, AgileCoder)	Assessor, Planner, Tutor and Checker roles	[21], [22]

Based on the supervisor's course slides [16], [23] and the DeepLearning.AI short courses he recommended [24].

LITERATURE REVIEW

Frameworks for Critical Thinking

Framework	Year	Core idea	Role in this project	Limitation	Ref
Paul's 6 Socratic question levels	-	Clarify, assumptions, evidence, viewpoints, implications, question the question	The ladder: what the Tutor asks next	Says what to ask, not how good the answer is	[1]
Paul-Elder 9 intellectual standards	-	Clarity, accuracy, precision, relevance, depth, breadth, logic, significance, fairness	The Assessor marks each as 1 or 0	Does not say which type of reasoning was used	[1]
Paul-Elder 8 elements of thought	-	Purpose, question, information, inference, concepts, assumptions, implications, point of view	Prerequisites before reasoning (planned)	Links to reasoning types still open	[1]
Revised Bloom's taxonomy	2001	Knowledge dimension: factual, conceptual, procedural, metacognitive	Types of knowledge used (planned)	No causal or relational type	[2]
SOLO taxonomy	1982	Prestructural to extended abstract outcomes	Complexity: how many knowledge types are joined	Needs adapting to dialogue	[3]

LITERATURE REVIEW

Socratic Questioning and Socratic AI

Study	Year	Setting	Main finding	Limitation	Ref
Sahamid	2016	Classroom action research	Socratic questioning built critical thinking	Small, one class	[4]
Makhene	2019	Nursing education	Socratic inquiry helps critical thinking	Needs skilled teachers' time	[5]
Ho, Chen and Li	2023	Healthcare students	Socratic method improved thinking skills	Human tutor, does not scale	[6]
Degen and Asanov	2025	65 pre-service teachers, Socratic AI vs plain chatbot	More support for critical, reflective thinking; argues for multi-agent systems	Self-reported, not a direct measure	[7]
Fan	2025	Socratic scaffolding through AI	Structured scaffolding deepened inquiry	No link to reasoning constructs	[8]
Lee, Ang and Lai	2025	Bloom labels on chatbot dialogues	Dialogue logs reveal thinking processes	Descriptive only	[9]
Lee et al. (CHI)	2025	319 knowledge workers	More AI trust, less critical thinking	Survey, self-reported	[10]

LITERATURE REVIEW

Existing Socratic AI Tutors

Tool	Refuses answers	Question ladder	Framework assessment	Goal, progress, end	Reasoning and knowledge types
Online Gemini Gem "Socratic Tutor" (Test #1)	Yes	No	No (informal points)	No	No
Project prompt v1 (Test #2)	Yes	Partly	Partly (hidden 1 to 3 score)	No	No
Khanmigo, custom GPTs *	Yes	Not shown	Not shown	Partly	No
Socratic AI (this project)	**Yes**	**Yes, 6 levels**	**Yes, 9 standards**	**Yes**	**Planned**

** From public descriptions, not tested in this project.*

No tested tool combines a question ladder, framework assessment and a clear session, and none shows which type of reasoning a learner used.

LITERATURE REVIEW

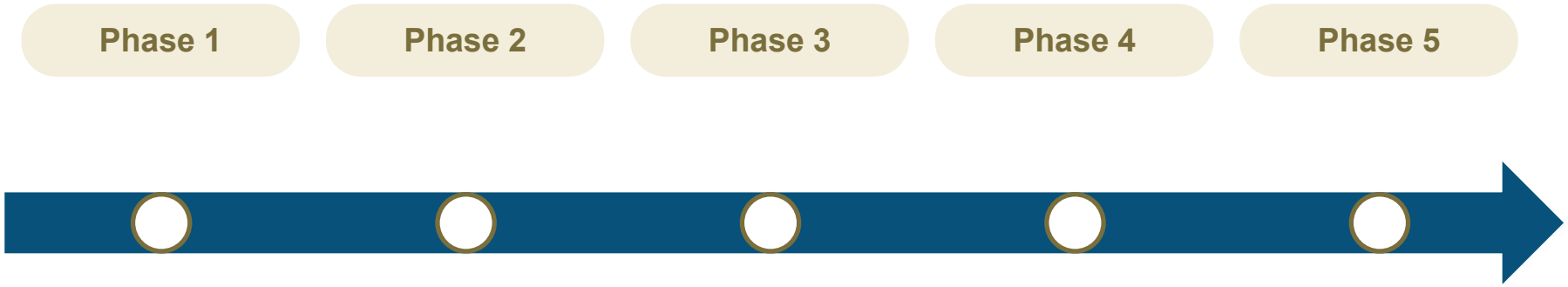
RESEARCH GAP

- AI use can reduce critical thinking effort, and current tutors are mostly answer machines.
- Socratic AI tutors are promising, but their questions have no principled ladder.
- Reasoning is rarely measured against a defined framework, and evidence is mostly self-reported.
- No tool shows the learner which type of reasoning or knowledge they used.

JUSTIFICATION

- A multi-agent tutor where the Tutor follows Paul's six levels, so questions climb in a set order.
- A separate Assessor marks every answer against the nine standards, with evidence, so it can be checked against human markers.
- Plain-code rules guard every AI decision, so steps are fair and testable ("agents decide, rules guard").
- Next, new agents add reasoning type and knowledge type, measured over many questions.

PROPOSED METHODOLOGY



Early testing
Two Gemini Gem tutors tested; problems written down

Requirements
9 design requirements R1 to R9 from the tests

Build
4 agents, rules, 3 cases, simulations, web app

Test and fix
Real sessions read turn by turn; problems fixed

Evaluate
Requirements, Assessor vs humans, learners

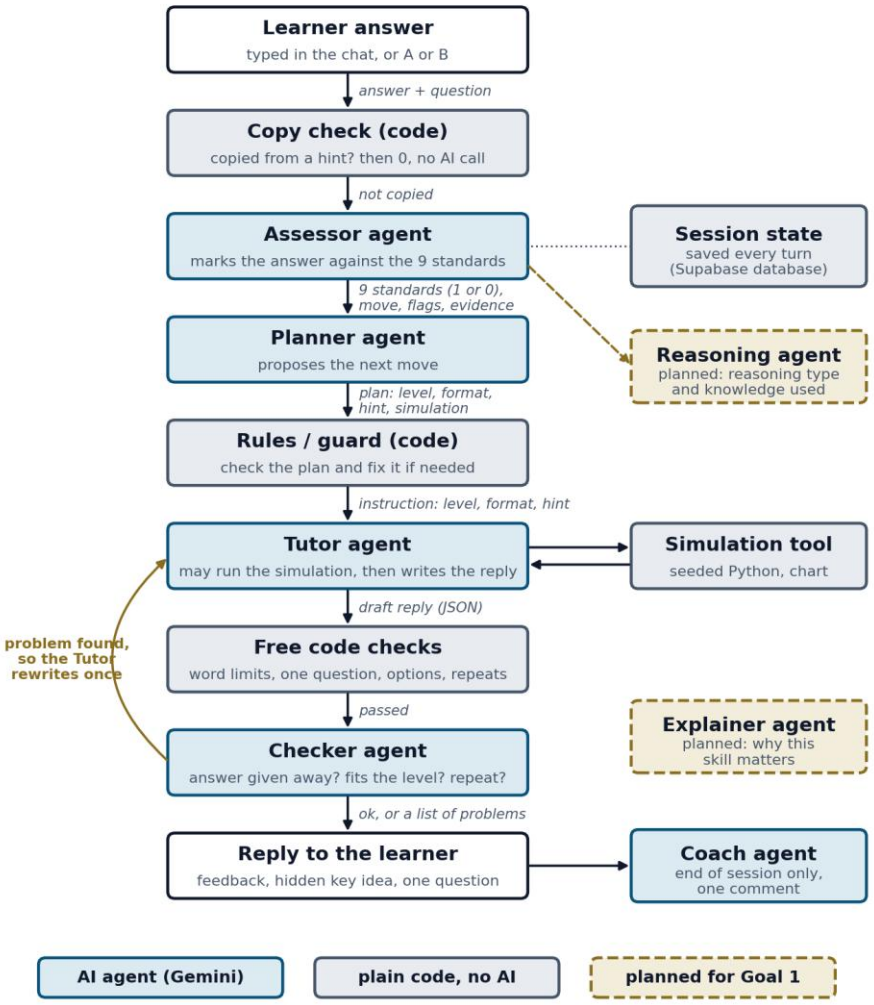
R1 Clear mission	R2 Fixed rounds, progress	R3 Remember, never re-ask
R4 Feedback and teaching	R5 Notice frustration	R6 Vary formats
R7 Separate assessor	R8 Learner stats	R9 Stable topic

Software
- Python 3.13, FastAPI, LangGraph
- Gemini 3.5 Flash Lite (free tier)
- HTML and JavaScript front end
- Supabase database, Render hosting (free)
- pytest: 94 automatic tests

Hardware
- A normal laptop; the AI runs in the cloud, so no GPU is needed

PROPOSED METHODOLOGY

System Architecture: Agents Decide, Rules Guard

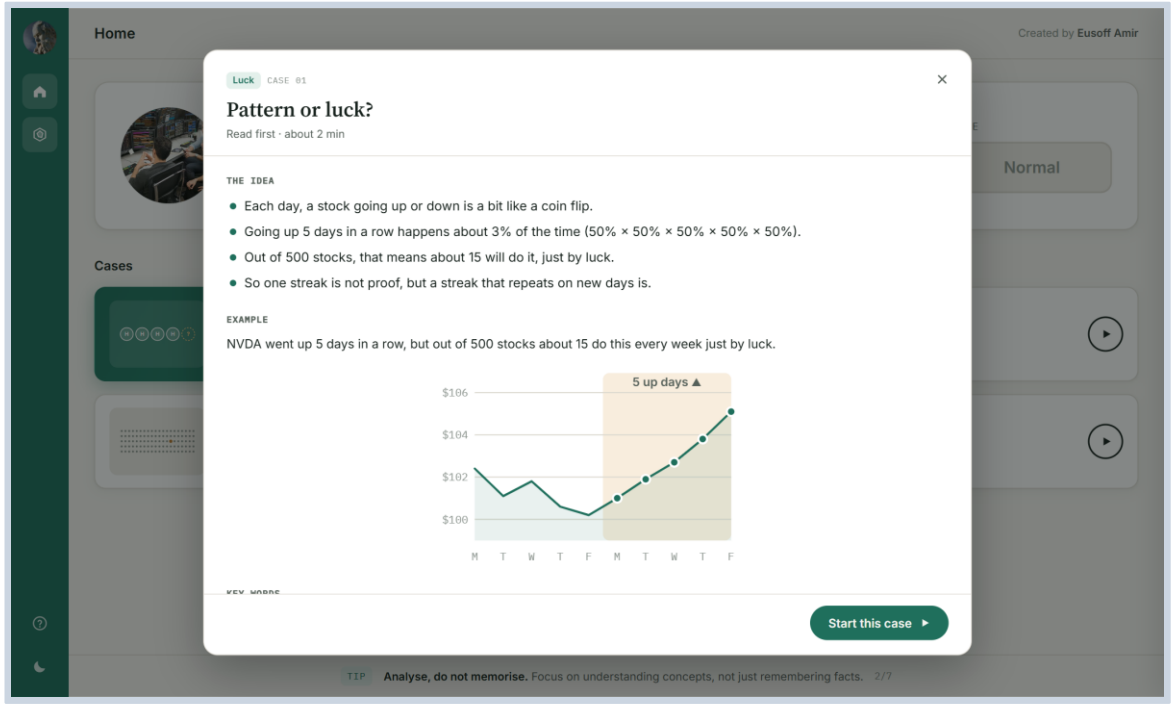


1. **Assessor:** marks each answer, 9 standards as 1 or 0, with an evidence quote
2. **Planner:** proposes the next level, question format, hint and simulation
3. **Rules (code):** a frustrated learner always gets an easy A or B question; no level up after a weak answer
4. **Tutor:** short feedback, a hidden hint, one question; may run a quant simulation
5. **Checker:** stops replies that give away the answer or repeat a question
6. **Coach:** one comment at the end on how the learner answered

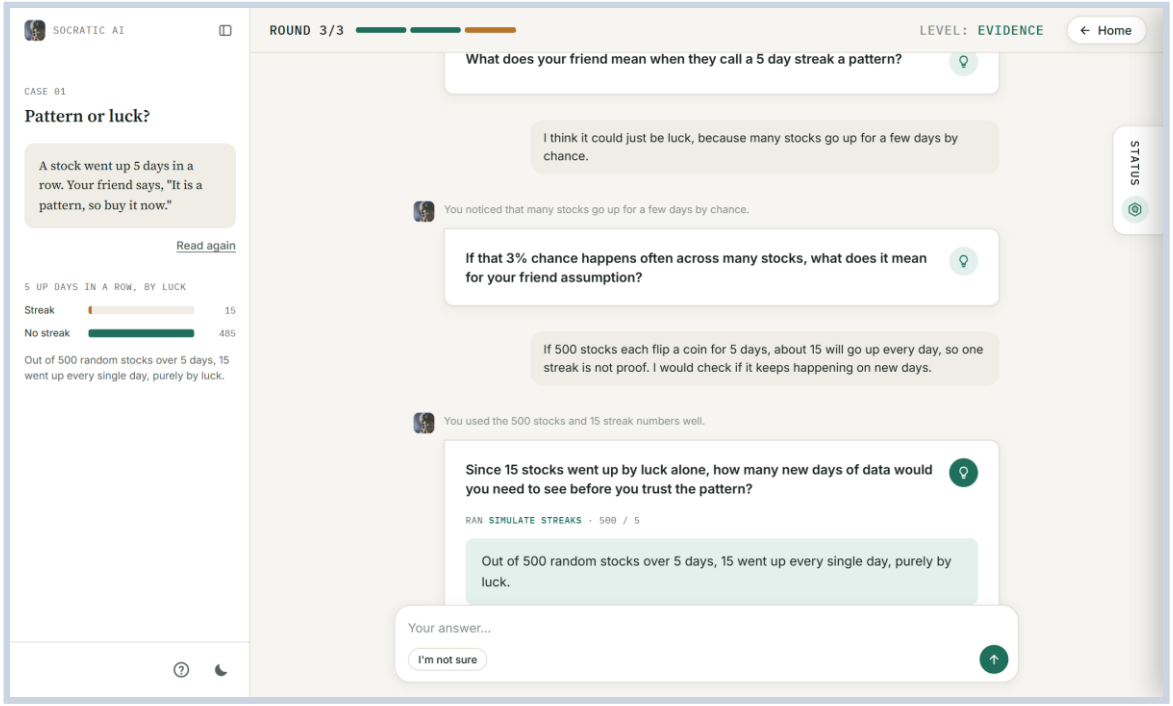
Each agent gets a small JSON payload built by code and must reply in a fixed JSON shape. About 25 AI calls per 6-round session.

PROPOSED METHODOLOGY

Session Design (R1 to R9)



1. A short reading and a mission before each case

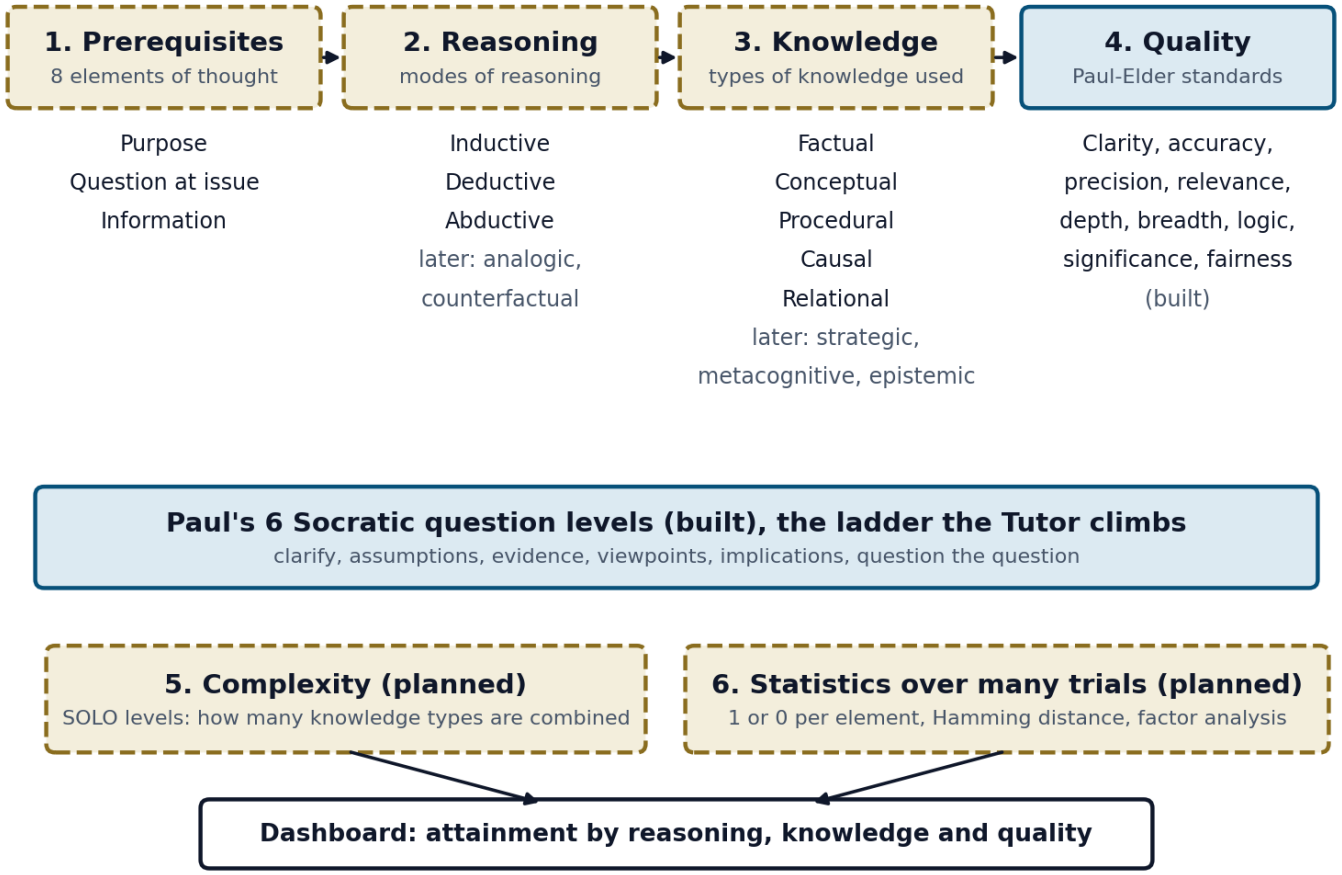


2. Feedback, one question, a hidden hint, and a simulation result

3, 6 or no-limit rounds | **Round and level shown** | **4 question formats** | **"I'm not sure" button** | **Status panel** | **Tutorial**

PROPOSED METHODOLOGY

Next Stage (Goal 1): Reasoning and Knowledge Layer



- Knowledge layer: each case note tagged with its type of knowledge; an agent drafts a knowledge graph
- Reasoning agent: 1 or 0 for inductive, deductive and abductive reasoning and each knowledge type, with evidence
- Question bank: several questions per skill, so there are enough trials
- Statistics: share of 1s and Hamming distance per learner [14]; factor analysis across learners
- Explainer agent: a plain scenario on why a skill matters

Example: "All the regional banks fell, not just CIMB" is inductive reasoning on relational knowledge.

PROPOSED METHODOLOGY

Evaluation Plan

1 Requirement check

Objective 1

Run the same scenarios on the baseline Gem tutors and on Socratic AI, and mark R1 to R9 as met or not.

2 Assessor vs human markers

Objective 2

Mark 61 test answers by hand with a marking guide. Compare using percent agreement and Cohen's kappa per standard (parent grant target 0.8).

3 With and without agents

All objectives

Switch the Planner, simulations or Checker off, and compare the same sessions.

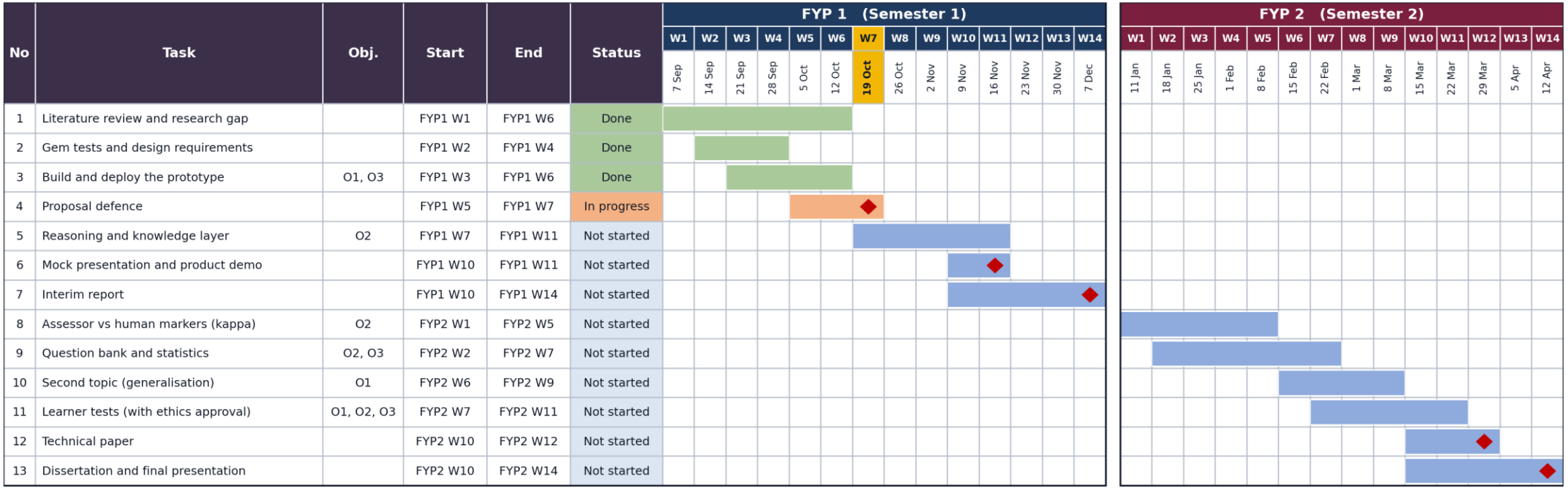
4 Learner experience

Objective 3

A short questionnaire with Dr Ho's students, after ethics approval.

GANTT CHART & MILESTONES

No	Task	Obj.	Start	End	Status
1	Literature review and research gap		FYP1 W1	FYP1 W6	Done
2	Gem tests and design requirements		FYP1 W2	FYP1 W4	Done
3	Build and deploy the prototype	O1, O3	FYP1 W3	FYP1 W6	Done
4	Proposal defence		FYP1 W5	FYP1 W7	In progress
5	Reasoning and knowledge layer	O2	FYP1 W7	FYP1 W11	Not started
6	Mock presentation and product demo		FYP1 W10	FYP1 W11	Not started
7	Interim report		FYP1 W10	FYP1 W14	Not started
8	Assessor vs human markers (kappa)	O2	FYP2 W1	FYP2 W5	Not started
9	Question bank and statistics	O2, O3	FYP2 W2	FYP2 W7	Not started
10	Second topic (generalisation)	O1	FYP2 W6	FYP2 W9	Not started
11	Learner tests (with ethics approval)	O1, O2, O3	FYP2 W7	FYP2 W11	Not started
12	Technical paper		FYP2 W10	FYP2 W12	Not started
13	Dissertation and final presentation		FYP2 W10	FYP2 W14	Not started

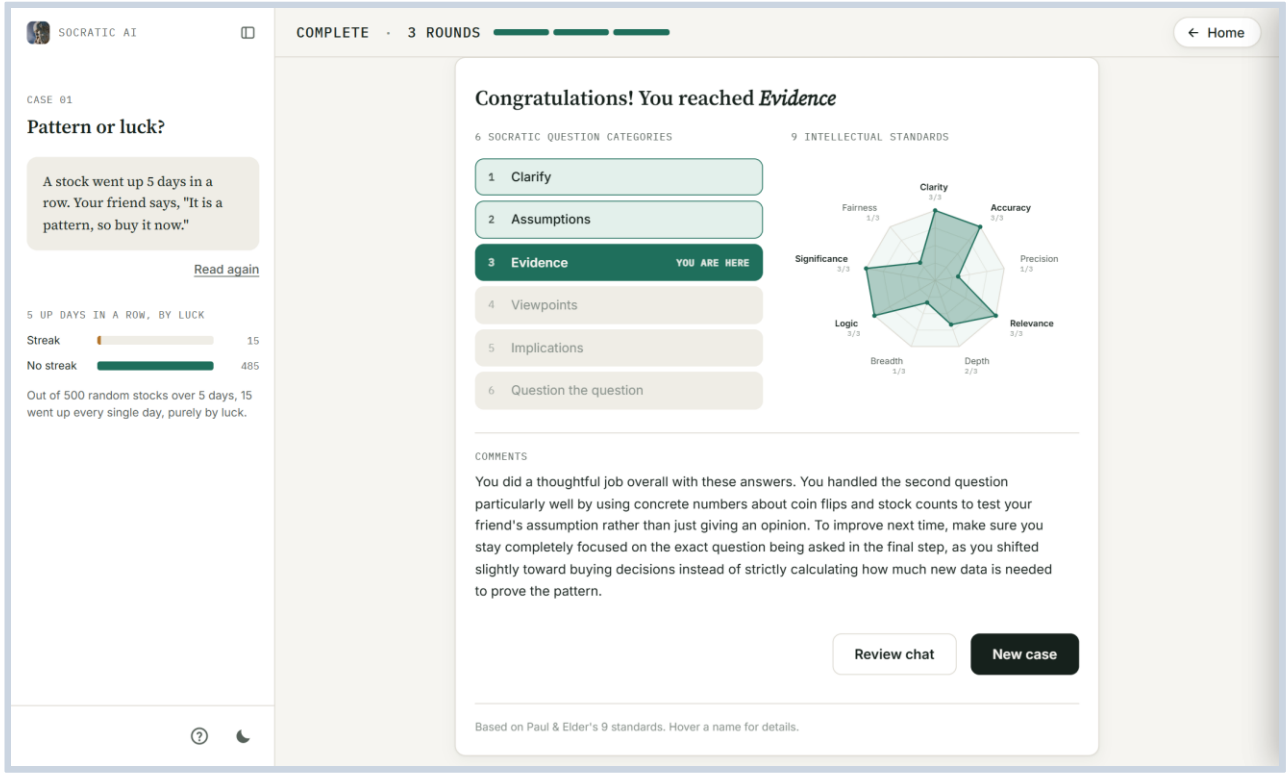


O1 to O3 are the three objectives (teach, assess, adapt)

- **21 Oct 2026 (FYP1 W7)** — Proposal defence
- **16 Nov 2026 (FYP1 W11)** — Product demo, mock presentation
- **7 Dec 2026 (FYP1 W14)** — Interim report
- **29 Mar 2027 (FYP2 W12)** — Technical paper
- **12 Apr 2027 (FYP2 W14)** — Dissertation, final presentation

PRELIMINARY RESULTS

A Real 3-Round Session (Gemini)



End profile: level reached, the 9 standards chart and the Coach comment

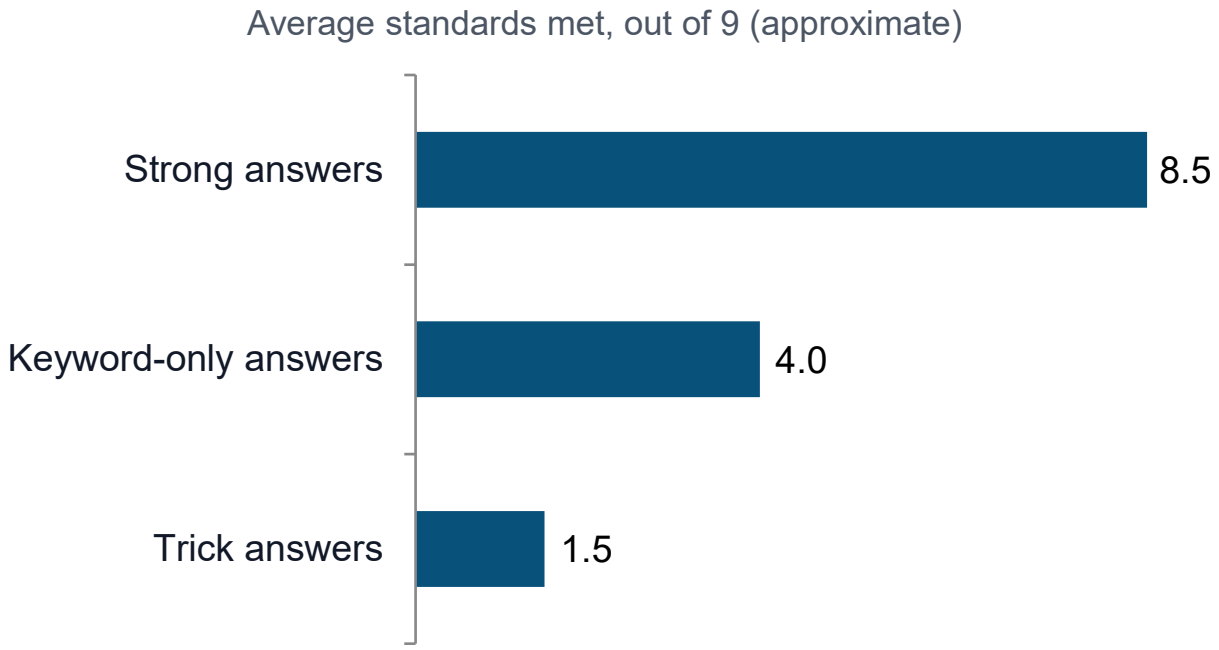
0 times the answer was given away, even when asked to ignore the rules

5 of 5 problems found in testing were fixed

~24 AI calls per session, inside the free tier

PRELIMINARY RESULTS

Assessor on a 61-Answer Test Set



37 real answers from test sessions and 24 constructed answers. Not yet compared with human markers, so this is not a validation yet.

Problems found and fixed

- The welcome said there was no round limit: now written by code
- The same question asked again in new words: the Checker now compares questions
- The Planner's simulation was ignored: the rules now run it
- Levels rose too fast: now 2 good answers in a row, or 1 strong answer
- The Assessor was generous: strict rules per standard; standards met fell from 24 to 16 on 12 real answers

Next: keyword-only answers still score about 4 of 9, so the Assessor prompt is tightened again.

CONCLUSIONS

Existing Socratic AI tutors have no question ladder, no measurement and no adaptation. Socratic AI is a working multi-agent tutor that follows Paul's six levels, marks every answer against the nine standards, adapts to the learner, and never gives the answer away. Next, it will show which type of reasoning and knowledge a learner used, and be checked against human markers.

Take-home message

This is a reasoning trainer, not an answer machine, built in a new way: many prompted agents working together, guarded by code.

REFERENCES

[1] R. Paul and L. Elder, The Thinker's Guide to the Art of Socratic Questioning. Foundation for Critical Thinking, 2006; and The Miniature Guide to Critical Thinking Concepts and Tools, 8th ed. Rowman & Littlefield, 2019.

[2] L. W. Anderson and D. R. Krathwohl, Eds., A Taxonomy for Learning, Teaching, and Assessing. Longman, 2001.

[3] J. B. Biggs and K. F. Collis, Evaluating the Quality of Learning: The SOLO Taxonomy. Academic Press, 1982.

[4] H. Sahamid, "Developing critical thinking through Socratic questioning," Int. J. Educ. Literacy Stud., vol. 4, no. 3, pp. 62-72, 2016.

[5] A. Makhene, "The use of Socratic inquiry to facilitate critical thinking in nursing education," Health SA Gesondheid, vol. 24, a1224, 2019.

[6] Y.-R. Ho, B.-Y. Chen and C.-M. Li, "Thinking more wisely: Using the Socratic method to develop critical thinking skills amongst healthcare students," BMC Med. Educ., vol. 23, 173, 2023.

[7] P.-B. Degen and I. Asanov, "Beyond automation: Socratic AI, epistemic agency, and ... multi-agent learning architectures," arXiv:2508.05116, 2025.

[8] F. Fan, "Socratic scaffolding in AI education: A framework for critical thinking," Regional Educ. Technol. J., vol. 8, no. 1, 2025.

[9] K. W. Lee, Y. S. Ang and J. W. Lai, "Learning analytics with scalable Bloom's taxonomy labeling of Socratic chatbot dialogues," Computers, vol. 14, no. 12, 555, 2025.

[10] H.-P. Lee et al., "The impact of generative AI on critical thinking," in Proc. CHI '25, Art. 1121, 2025.

[11] A. Sarkar, "How to stop AI from killing your critical thinking," TEDAI Vienna, 2025.

[12] L. Brand, "A practical guide to agentic AI in higher education," QS, 11 Sep. 2026.

REFERENCES

[13] L. Yan et al., "Practical and ethical challenges of large language models in education," Br. J. Educ. Technol., vol. 55, no. 1, pp. 90-112, 2023.

[14] R. W. Hamming, "Error detecting and error correcting codes," Bell Syst. Tech. J., vol. 29, no. 2, pp. 147-160, 1950.

[15] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," NeurIPS, vol. 33, pp. 9459-9474, 2020.

[16] T. W. Ho, Foundations of Modern Deep Neural Networks, Module 8: Towards Agentic AI [Course slides]. CAPE, Universiti Teknologi PETRONAS, 2025.

[17] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," NeurIPS, vol. 35, pp. 24824-24837, 2022.

[18] S. Yao et al., "ReAct: Synergizing reasoning and acting in language models," in Proc. ICLR, 2023.

[19] Y. Gao et al., "Retrieval-augmented generation for large language models: A survey," arXiv:2312.10997, 2023.

[20] Anthropic, "Introducing the Model Context Protocol," 25 Nov. 2024. [Online]. Available: https://www.anthropic.com/news/model-context-protocol

[21] S. Hong et al., "MetaGPT: Meta programming for a multi-agent collaborative framework," in Proc. ICLR, 2024.

[22] M. H. Nguyen, T. P. Chau, P. X. Nguyen and N. D. Q. Bui, "AgileCoder: Dynamic collaborative agents for software development based on agile methodology," arXiv:2406.11912, 2024.

[23] T. W. Ho, Foundations of Modern Deep Neural Networks, Module 7: Foundation Models and Generative AI [Course slides]. CAPE, Universiti Teknologi PETRONAS, 2025.

[24] DeepLearning.AI, short courses: Knowledge Graphs for RAG; Building and Evaluating Advanced RAG; Agentic Knowledge Graph Construction; Long-Term Agentic Memory with LangGraph; Building Agentic RAG with LlamaIndex; Retrieval Augmented Generation. [Online]. Available: https://learn.deeplearning.ai

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Q & A